Prompt for Graph-based Extraction.

Entity Extraction:

Previous messages:
`{previous messages}`
Current message:
`{current message}`
Given the above conversation, extract entity nodes from the current message that are explicitly or implicitly mentioned. Note that ALWAYS extract the speaker/actor as the first node and extract other significant entities, concepts, or actors mentioned in the current message.

INSTRUCTIONS:
1. ALWAYS extract the speaker/actor as the first node. The speaker is the part before the colon in each line of dialogue.
2. Extract other significant entities, concepts, or actors mentioned in the current message.
3. DO NOT create nodes for relationships or actions.
4. DO NOT create nodes for temporal information like dates, times or years (these will be added to edges later).
5. Be as explicit as possible in your node names, using full names.
6. DO NOT extract entities mentioned only.

Relation Extraction:

Previous messages:
`{previous messages}`
Current message:
`{current message}`
Entities:
`{entities}`
Given the above messages and entities, extract all facts pertaining to the listed entities from the current message. Note that extract facts only between the provided entities and each fact should represent a clear relationship between two DISTINCT nodes.

INSTRUCTIONS:
1. Extract facts only between the provided entities.
2. Each fact should represent a clear relationship between two DISTINCT nodes.
3. The relation_type should be a concise, all-caps description of the fact (e.g., LOVES, IS_FRIENDS_WITH, WORKS_FOR).
4. Provide a more detailed fact containing all relevant information.
5. Consider temporal aspects of relationships when relevant.

Figure 12: A sample prompt for graph-based extraction.

A PROMPT TEMPLATES

This section details several prompt templates used in our evaluation. A sample prompt for graph-based information extraction is shown in Figure 12, and the prompt for post-processing answer simplification is provided in Figure 13.

B EXPERIMENT DETAILS

B.1 Dataset Details

In this section, we provide a detailed breakdown of the specific evaluation categories for both the *LOCOMO* and *LONGMEMEVAL* benchmarks.
LOCOMO evaluates agent memory capabilities across diverse scenarios. The specific task categories are defined as follows:

- **Single-hop** questions require answers based on a single session.
- **Multi-hop** questions require synthesizing information from multiple different sessions.
- **Temporal reasoning** questions require temporal reasoning and capturing time-related data cues.
- **Open-domain knowledge** questions require integrating provided information with external knowledge such as commonsense or world facts.

Prompt for Answer Simplification.

Your task is to act as an answer simplifier. I will give you a question and a full-sentence answer. You must reduce the answer to its most critical component.
Follow these rules:
1. **Extract the Core Information:** Identify the primary piece of information that directly answers the question.
2. **Remove Extraneous Phrases:** Eliminate phrases like "Based on the information provided...", "The answer is...", "As per the document...", etc.
3. **Omit Explanations:** Do not include any justifications, reasoning, or additional context from the original answer.
4. **Be Concise:** The output should be the shortest possible string that still accurately answers the question.

Example:
- Question: "What degree did I graduate with?"
- Original Answer: "Based on the information provided, you graduated with a degree in Business Administration."
- Simplified Answer: "Business Administration"

Here is the question and answer to simplify:
- Question: `{question}`
- Original Answer: `{answer}`
- Simplified Answer:

Figure 13: Prompt for answer simplification.

LONGMEMEVAL categorizes memory tasks to assess the following aspects:

- **Information Extraction (IE):** Ability to recall specific information from extensive interactive histories, including the details mentioned by either the user (**single-session-user**) or the assistant (**single-session-assistant**), and whether the model can utilize the user information to generate a personalized response (**single-session-preference**).

- **Multi-Session Reasoning (MR):** Ability to synthesize the information across multiple history sessions to answer complex questions that involve aggregation and comparison.
- **Knowledge Updates (KU):** Ability to recognize the changes in the user's personal information and update the knowledge of the user dynamically over time.
- **Temporal Reasoning (TR):** Awareness of the temporal aspects of user information, including both explicit time mentions and timestamp metadata in the interactions.

B.2 Dataset Variant Construction

LONGMEMEVAL is well-suited for controlled evaluation: it features a configurable context length enabled by a *session pool*, which supplies topically similar history sessions without introducing conflicting information, and each sample is associated with a single question, making the corresponding ground-truth evidence clearly defined. Based on these properties, we construct a set of variants of the original dataset (LONGMEMEVAL$_S$) to evaluate **context scalability** and **position sensitivity**.

For **context scalability**, we construct variants at 50%, 150%, and 200% of the default context length. Specifically, for the 50% variant, we prune half of the history dialogues at session-level granularity while retaining the ground-truth session; for the 150% and 200% variants, we append additional sessions sampled from the session pool to the end of the original conversation history.

For **position sensitivity**, we generate position-varying variants of LONGMEMEVAL$_S$ by relocating the ground-truth session. Specifically, we extract the ground-truth session from its original position and re-insert it into one of three equally partitioned segments of the conversation: the first 1/3 (*Early*), the middle 1/3 (*Middle*), and the final 1/3 (*Late*).

B.3 Detailed Results

In this section, we present detailed results for Exp.3 Context scalability analysis and Exp.4 Position sensitivity analysis using Qwen2.5-7B-Instruct. Note that Zep is excluded as it cannot finish on LONGMEMEVAL within two days. Results are shown in Table 9 and Table 10.

C DETAILS OF OUR NEWLY DESIGNED METHOD

This section details the complete workflow of our newly designed method and the algorithm is provided in Algorithm 1.

As illustrated in Figure 11, new messages are first ingested into short-term memory and managed via a FIFO queue. When the short-term memory reaches capacity, the oldest messages are partitioned into segments based on semantic similarity and transferred to mid-term memory. Within this tier, we maintain a memory tree in which each leaf node represents a segment—capturing a generated summary of its constituent messages—while parent nodes provide aggregated summaries of their respective children. This segment-level granularity significantly reduces token overhead compared to turn-level processing. For each segment leaf node, we compute a *heat score* based on access frequency and recency; segments with high heat scores are promoted to long-term memory.

During the information retrieval stage, we conduct independent retrieval from each of the three storage tiers. Short-term memory is retrieved in its entirety to maintain context continuity. For mid-term memory, we employ a dual-mode retrieval mechanism: high-level node semantics are matched via flat vector-based similarity search, while raw messages are retrieved through a beam search that traverses from the root node, selecting the top-k most similar nodes at each level and ultimately reaching the raw messages stored as children of the segment leaf nodes. Long-term memory is accessed via standard vector-based similarity retrieval.

Algorithm 1: Our Newly Designed Method

input : Current query q, Short-term memory $\mathcal{M}_S$, Mid-term memory $\mathcal{M}_M$, Long-term memory tree $\mathcal{M}_L$, short-term capacity threshold τ, heat threshold θ, beam width k
output : Updated memories $(\mathcal{M}'_S, \mathcal{M}'_M, \mathcal{M}'_L)$, Response r

```
// (1) Information Retrieval
C_S ← M_S ;          // retrieve entire short-term memory
C_M_flat ← VectorSearch(M_M.high-level-nodes, q);
C_M_beam ← BeamSearch(M_M.tree, q, k);
C_L ← VectorSearch(M_L, q);
C ← C_S ∪ C_M_flat ∪ C_M_beam ∪ C_L ; // aggregate retrieved context
// (2) Response Generation
r ← LLM_Generate(q, C);
// (3) Memory Ingestion
M'_S ← Enqueue(M_S, q, r) ;      // ingest new messages via short-term FIFO queue
if |M'_S| > τ then
    M_old ← DequeueOldestHalf(M'_S);
    S_new ← SegmentBySemanticSimilarity(M_old) ; // partition into segments
    M'_M ← UpdateMemoryTree(M_M, S_new) ;        // leaf: segment, parents: aggregated summary
else
    M'_M ← M_M;
M'_L ← M_L;
foreach leaf node n ∈ M'_M.leaves do
    score ← ComputeHeatScore(n.freq, n.recency);
    if score > θ then
        M'_L ← M'_L ∪ {n} ;  // promote high-heat segments
return (M'_S, M'_M, M'_L, r)
```

D MORE LESSONS AND OPPORTUNITIES

Lessons:

▶ **L4.** Well-designed memory frameworks decouple reasoning from model scale, enabling smaller LLMs to effectively handle complex queries such as those involving temporal dependencies. For instance, employing multi-step reasoning or specialized components within the framework allows smaller LLMs to address complex tasks more effectively.

▶ **L5.** Instead of destructive updates, memory systems should adopt non-destructive strategies that preserve historical information while annotating its validity, enabling future reuse and preventing the loss of potentially useful knowledge. In other words, instead of deleting old information, systems should keep it and mark its status.

Opportunities:

Table 9: Comparison of methods on LONGMEMEVAL at various context scales.

Method	Scale	IE		MR		TR		KU		Overall	
		F1	BLEU-1	F1	BLEU-1	F1	BLEU-1	F1	BLEU-1	F1	BLEU-1
A-MEM	50%	41.54	35.29	17.82	15.63	20.67	16.12	24.24	20.98	26.98	22.72
	100%	39.70	33.23	16.61	14.32	19.37	15.54	25.59	21.63	25.95	21.84
	150%	37.92	31.45	15.48	13.92	20.08	17.51	21.43	18.91	24.63	21.11
	200%	36.68	30.12	16.81	14.15	20.94	16.39	21.87	18.42	24.89	20.44
MemoryBank	50%	42.04	34.88	19.06	15.65	23.62	18.08	32.93	26.53	29.61	24.11
	100%	41.19	29.39	19.41	16.42	17.80	13.15	31.51	27.31	27.65	23.08
	150%	42.37	34.52	15.62	12.93	18.57	12.56	27.64	22.06	26.63	21.03
	200%	41.56	33.74	16.69	13.35	17.43	12.83	31.02	24.52	26.85	21.32
MemGPT	50%	47.61	43.47	26.89	25.64	28.60	22.79	40.04	36.07	35.86	32.07
	100%	45.08	37.89	18.61	16.71	21.65	14.81	28.09	24.85	29.17	24.08
	150%	42.89	36.83	20.45	19.12	22.48	22.01	23.37	16.46	28.45	25.01
	200%	43.75	36.73	17.77	14.95	25.97	20.66	24.79	21.94	29.15	24.31
Mem0	50%	45.43	37.99	27.03	25.14	28.10	20.93	45.74	39.95	35.97	30.56
	100%	39.17	39.10	23.02	20.72	28.86	21.19	41.28	36.37	32.44	29.58
	150%	36.88	31.09	19.59	16.91	27.50	20.55	44.32	38.84	30.94	26.37
	200%	40.60	34.21	23.37	20.70	29.20	21.92	38.76	32.78	32.69	27.20
Mem0^g	50%	34.06	32.24	26.01	23.83	28.65	29.73	46.31	48.06	32.38	31.81
	100%	32.37	30.86	24.42	22.04	29.16	21.60	40.43	35.51	29.66	26.83
	150%	29.53	24.05	19.71	16.61	26.29	18.54	33.98	30.37	26.75	21.12
	200%	32.31	26.42	21.44	19.25	29.33	20.84	39.56	34.97	29.75	24.44
MemoChat	50%	12.49	9.01	9.63	7.57	19.37	12.78	5.92	4.18	12.53	8.88
	100%	10.36	6.05	12.27	10.09	17.49	11.35	6.51	4.28	12.16	8.26
	150%	12.52	7.92	11.94	10.44	16.85	9.96	3.29	1.75	12.08	8.17
	200%	11.08	6.72	10.70	9.35	19.04	12.44	6.48	4.11	12.38	8.53
MemTree	50%	50.42	41.71	26.59	24.26	29.19	21.93	51.66	46.73	38.63	32.59
	100%	53.30	44.72	21.96	19.88	29.97	21.77	41.49	38.56	36.92	31.05
	150%	46.67	38.81	26.01	24.00	25.38	17.96	43.13	38.32	34.96	29.25
	200%	52.98	44.22	27.04	24.71	25.69	18.41	47.34	42.72	37.94	31.93
MemoryOS	50%	51.78	43.82	22.53	20.77	23.72	17.81	39.09	35.26	34.56	29.44
	100%	49.91	43.15	19.35	17.80	26.34	21.61	30.62	27.97	32.50	28.31
	150%	50.32	42.78	21.13	19.95	22.93	17.73	34.81	31.16	32.85	28.23
	200%	44.55	36.92	21.30	19.22	21.93	17.70	38.98	36.01	31.48	26.79
MemOS	50%	50.35	38.11	22.97	19.16	21.04	14.99	37.38	33.18	33.24	26.15
	100%	49.46	40.52	22.81	19.72	21.34	15.65	33.99	27.75	32.48	26.38
	150%	43.33	32.93	20.76	17.52	21.89	15.14	22.68	17.05	28.40	21.62
	200%	37.08	28.68	20.12	16.83	24.84	16.33	27.76	20.76	27.86	21.01

▶ **O4.** In complex query scenarios, relying on a single fixed retrieval strategy can be brittle and may lead to task failure. Different queries often require different retrieval granularity and mechanisms. However, existing memory systems do not consider this issue, so designing a retrieval routing planner that dynamically selects and adapts retrieval strategies to diverse memory-query contexts is an interesting research problem.

▶ **O5.** Existing memory benchmarks, such as LOCOMO and LONGMEMEVAL, evaluate memory methods on pre-collected, static interaction histories, which do not reflect the continuous and evolving nature of real-world memory. They also fail to capture key properties of human memory such as the preference for recent updates and the strengthening of repeatedly mentioned information. Moreover, despite the inclusion of sparse visual elements in LOCOMO, these benchmarks remain strictly text-centric. Developing more challenging, interaction-driven, and comprehensive multimodal benchmarks that better reflect realistic memory formation and usage scenarios is a meaningful research direction.

Table 10: Positional Robustness Evaluation on the LONGMEMEVAL Dataset

Method	IE						MR		TR		KU		Overall	
	user		assistant		preference									
	F1	BLEU-1	F1	BLEU-1	F1	BLEU-1	F1	BLEU-1	F1	BLEU-1	F1	BLEU-1	F1	BLEU-1
Early Evidence Sessions														
A-MEM	41.62	37.81	42.49	36.54	8.74	0.73	10.48	8.15	20.31	17.65	28.61	24.59	23.76	20.13
MemoryBank	46.01	38.36	48.08	39.74	17.09	5.71	16.95	13.32	21.60	16.34	31.25	24.93	27.98	21.94
MemGPT	54.09	48.87	50.27	43.05	10.34	1.23	24.39	21.94	29.58	22.50	35.74	31.51	33.75	28.47
Mem0	63.86	57.57	33.98	27.04	10.93	0.40	25.13	22.65	29.78	21.66	34.98	30.81	33.46	27.71
Mem0^g	53.95	48.70	19.62	14.92	9.93	0.75	18.31	15.90	27.88	20.12	41.37	37.03	29.09	23.89
MemoChat	6.44	3.24	16.95	12.74	7.18	0.12	12.27	10.09	17.49	11.35	6.51	4.28	12.16	8.26
MemTree	52.85	48.92	49.26	37.66	9.31	0.18	26.18	23.72	27.54	20.39	40.83	36.40	34.13	28.49
MemoryOS	52.19	49.58	61.32	52.71	12.40	1.18	19.35	17.80	26.34	21.61	30.62	27.97	31.85	27.76
MemOS	60.65	50.63	43.00	36.78	11.20	0.16	19.85	17.02	22.12	16.45	25.37	19.02	29.10	23.09
Middle Evidence Sessions														
A-MEM	40.23	39.12	43.51	37.98	6.15	0.48	9.87	7.41	18.92	14.71	29.45	25.66	23.13	19.65
MemoryBank	41.89	35.15	52.13	44.49	15.44	5.16	16.35	13.43	25.15	19.73	32.12	26.78	28.68	23.21
MemGPT	49.83	45.29	48.63	41.72	10.56	2.07	24.38	22.01	26.77	20.33	35.85	31.82	32.25	27.36
Mem0	66.46	60.15	34.52	28.05	11.29	1.01	24.82	21.91	28.07	20.26	42.37	36.83	34.53	28.59
Mem0^g	48.88	43.93	17.02	12.76	10.52	0.81	19.48	17.06	27.70	19.83	45.99	40.50	29.10	23.76
MemoChat	9.43	6.96	19.37	15.06	7.71	0.14	13.08	10.98	16.48	10.75	3.42	1.33	12.35	8.66
MemTree	60.03	54.29	60.82	49.21	10.58	0.69	27.39	24.75	26.25	19.35	40.60	36.58	36.45	30.59
MemoryOS	51.27	47.13	63.14	52.26	10.90	1.07	20.39	18.05	23.80	18.45	35.21	32.32	32.15	27.27
MemOS	64.24	57.41	49.26	38.28	11.97	0.50	24.15	19.73	23.58	17.95	27.22	24.14	32.17	26.14
Late Evidence Sessions														
A-MEM	50.84	47.21	41.27	32.45	9.68	0.82	12.03	9.56	24.63	18.72	28.34	27.52	26.49	22.11
MemoryBank	45.57	38.11	50.63	43.63	14.64	5.14	15.60	12.31	26.21	19.54	27.57	21.96	28.35	22.43
MemGPT	54.13	49.02	57.29	50.27	10.56	1.31	23.01	20.69	29.67	23.30	34.57	30.91	34.03	29.09
Mem0	64.06	58.32	35.17	28.08	10.93	0.40	25.71	22.07	30.37	22.89	42.04	45.81	35.04	30.44
Mem0^g	63.48	59.40	17.91	14.34	11.01	0.84	24.31	25.28	28.61	20.06	38.62	35.76	31.66	27.61
MemoChat	8.59	6.09	20.12	15.46	8.78	0.22	11.15	9.61	23.33	16.16	4.79	2.79	13.90	9.89
MemTree	63.59	57.77	58.18	47.98	9.72	0.93	27.45	23.73	29.59	21.18	39.71	36.07	37.37	31.09
MemoryOS	49.13	44.71	63.44	53.51	13.58	2.54	22.18	19.69	24.26	18.13	38.39	34.85	33.14	27.90
MemOS	69.88	49.34	51.30	38.25	11.86	0.18	23.46	15.87	26.25	17.80	31.62	22.41	34.40	23.65